

ITA0612 – MACHINE LEARNING

CO4-AT3

COMPARATIVE ANALYSIS

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QUESTION: 1

Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

ANSWER:

1. Case-Based Learning (CBL)

CBL stores previous customer cases along with their recommendations.

Example:

A new customer has a profile similar to an existing customer who purchased a laptop, mouse, and keyboard. The system retrieves that similar case and recommends similar products.

Working:

1. Store previous customer cases.
2. Receive a new customer's data.
3. Find the most similar past cases.
4. Reuse or adapt their solutions/recommendations.

New Data: New cases can be added directly without retraining a global model.

Adaptability: Very high because the system can use recent cases and adapt recommendations based on similar previous experiences.

Computational Efficiency: Prediction can be expensive when the database is large because many stored cases may need to be compared.

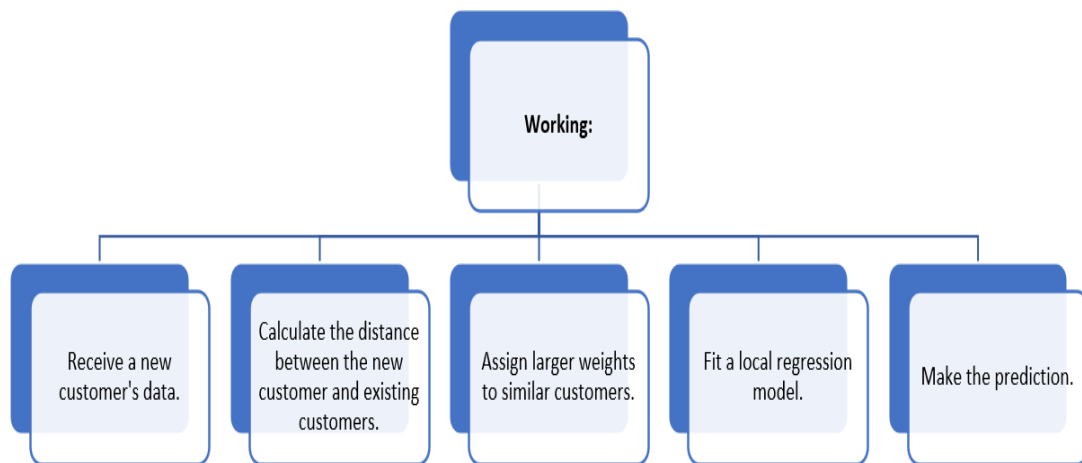
2. Locally Weighted Regression (LWR)

LWR gives higher weights to nearby/similar data points and lower weights to distant data points.

Example:

If customers similar to a new customer usually spend ₹5,000–₹7,000 on electronics, LWR uses these nearby customers more strongly to predict what products or price range should be recommended.

WORKING:



New Data: It can immediately use newly added data during prediction without retraining the entire model.

Adaptability: High because the model focuses on the local neighborhood and can capture changes in customer behavior.

Computational Efficiency: Prediction can be computationally expensive because distances and local models must be calculated for each new query, especially with a large dataset.

Customer	Age	Monthly Spending	Recommended Product
C1	20	3000	Smartphone
C2	22	3500	Smartphone
C3	25	5000	Laptop
C4	27	5500	Laptop
C5	30	7000	Laptop

Comparison

Factor	Case-Based Learning	Locally Weighted Regression
Main idea	Retrieves and reuses similar past cases	Gives higher weights to nearby data points
New data	New cases can be added directly without retraining	Uses new data during local prediction
Adaptability	Very high because it uses recent similar cases	Very high because it focuses on local patterns
Prediction	Based on similar stored cases	Based on weighted local regression
Computational efficiency	Can be costly for large case databases	Can be costly because distances and local regression are calculated
Best suited for	Recommendation systems and decision support	Numerical prediction and trend estimation
Interpretability	Easy because similar past cases can be shown	Moderate because predictions depend on weighted observations
Example	Recommend products based on similar customers	Predict customer spending based on nearby customer

Conclusion

Case-Based Learning is better when recommendations can be obtained by finding similar previous situations, while Locally Weighted Regression is better when we need a numerical prediction based on nearby observations. Both are highly adaptable to new data, but their prediction cost increases as the dataset becomes larger.

QUESTION: 2

Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

Locally Weighted Regression vs. Radial Basis Function Networks

Both Locally Weighted Regression (LWR) and Radial Basis Function (RBF) Networks are useful for function approximation, where the goal is to learn an unknown relationship between input and output.

Example Dataset

Temperature (°C)	Electricity Demand (kW)
20	120
22	135
24	150
26	170
28	195
30	225
32	260

Consider the following dataset representing **temperature (X)** and **electricity demand (Y)**:

Suppose we want to predict electricity demand when the temperature is **27°C**.

1. Locally Weighted Regression (LWR)

- LWR gives higher importance to data points close to the query point and lower importance to distant points.
- For 27°C, the observations at 26°C and 28°C receive high weights, while 20°C and 32°C receive lower weights.

Main characteristics:

- **Locality:** Very strong. Only nearby observations strongly influence the prediction.
- **Computational cost:** High during prediction because distances and weights must be calculated for each new query.
- **Smoothness:** Depends on the bandwidth. A small bandwidth produces highly local and potentially irregular predictions, while a larger bandwidth produces smoother predictions.
- **Training:** Very little or no explicit global training; computation is mainly performed when making predictions.

2. Radial Basis Function (RBF) Network

- An RBF network uses radial basis functions, commonly Gaussian functions, centered at selected points in the input space.
- For 27°C, RBF units centered near 26°C and 28°C will have stronger activation. The network combines these activations to produce the prediction.

Main characteristics:

- **Locality:** Local, because each RBF unit responds strongly to inputs near its center.
- **Computational cost:** Training can be more expensive because centers, widths, and weights need to be determined. However, prediction is generally faster after training.
- **Smoothness:** Usually produces smooth predictions because Gaussian basis functions overlap and combine gradually.
- **Training:** Requires a training phase to learn the RBF parameters.

Comparison

Factor	Locally Weighted Regression	Radial Basis Function Network
Basic idea	Fits a local regression around each query	Combines radial basis functions
Locality	Very high; directly uses nearby data	Local; determined by RBF centers and widths
Training cost	Low	Higher
Prediction cost	Higher for large datasets	Usually lower after training
Smoothness	Depends strongly on bandwidth	Generally smooth
New data	Easily incorporates new data	Usually requires updating or retraining parameters
Memory requirement	Requires storing training data	Stores learned centers widths and weights
Best use	Small or medium datasets with changing local patterns	Repeated predictions and nonlinear function approximation

Conclusion

LWR is more flexible and highly responsive to local changes, making it useful when new data arrives frequently. However, its prediction can be computationally expensive because it performs local calculations for each query.
